



澳門理工大學
Universidade Politécnica de Macau
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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis			
Date of Submission	2026-06-04		

Reviewer

Name	Qiu Huang	Date	2026-06-12
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

The dissertation presents efforts towards deep learning-based frameworks for early cancer screening using routine chest CT images, focusing on thyroid cancer and pulmonary nodules.

Knowledge Base and Literature Review.

The author demonstrates broad coverage of relevant fields including medical imaging, deep learning architectures (CNNs, Transformers, contrastive learning), and clinical applications in lung and thyroid cancer. It would be better if the review leans toward critical analysis instead of summarization.

Originality and Independence.

The dissertation proposes two main models: the DVT model for thyroid cancer screening using secondary transfer learning and Transformer, and the CL-IJLO framework for

longitudinal pulmonary nodule risk prediction using contrastive learning-based feature compression and a joint loss function.

Arguments and Reasoning.

The claimed superiority of the proposed models is supported by comparative experiments against several baselines, ablation studies, and multi-center validation.

Evaluation within the Field.

The thesis represents a solid engineering effort with competent application of deep learning to clinically relevant problems. It demonstrates awareness of field trends and makes a genuine attempt to address limitations of prior work.

Completeness and Coherence.

The thesis is structurally well-organized. The three-dimensional framework (spatial, temporal, modal) provides conceptual unity, and the conclusion revisits each contribution while acknowledging limitations.

Conclusion.

Recommended for oral defense.

Additional Comments.